

The slide features a white background with several decorative elements. On the left, there are four hexagons: a light blue one, a dark green one, a large teal one, and a small green one. On the right, there are overlapping blue geometric shapes, including triangles and polygons, creating a modern, abstract design.

# Car Market Trends Analysis with CarDekho Data

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# PROBLEM STATEMENT

- Used car buyers and sellers often lack a reliable, data-driven way to judge fair resale value. Prices depend on many interacting factors — fuel type, transmission, mileage, ownership history and vehicle age — which makes manual pricing inconsistent and confusing.
- This project analyzes 300+ real CarDekho used-car listings to uncover the key drivers of resale price and support smarter buying and selling decisions.



# Project Description

A hands-on data analytics project exploring real used-car listings from CarDekho to understand what actually drives resale price in the Indian second-hand car market.

Using Python and Pandas, a dataset of 301 cars (model years 2003-2018) was cleaned and analyzed across fuel type, transmission, seller type, mileage, ownership history and vehicle age to surface pricing patterns buyers, sellers and dealers can act on.

# WHO ARE THE END USERS?



- Used car buyers who want a fair, data-backed sense of a vehicle's worth before purchasing
- Individual sellers deciding how to price their car to sell quickly and fairly
- Car dealerships and resale platforms (e.g. CarDekho) refining pricing recommendations
- Loan and insurance providers estimating vehicle valuation
- Analysts and students studying trends in the automotive resale market



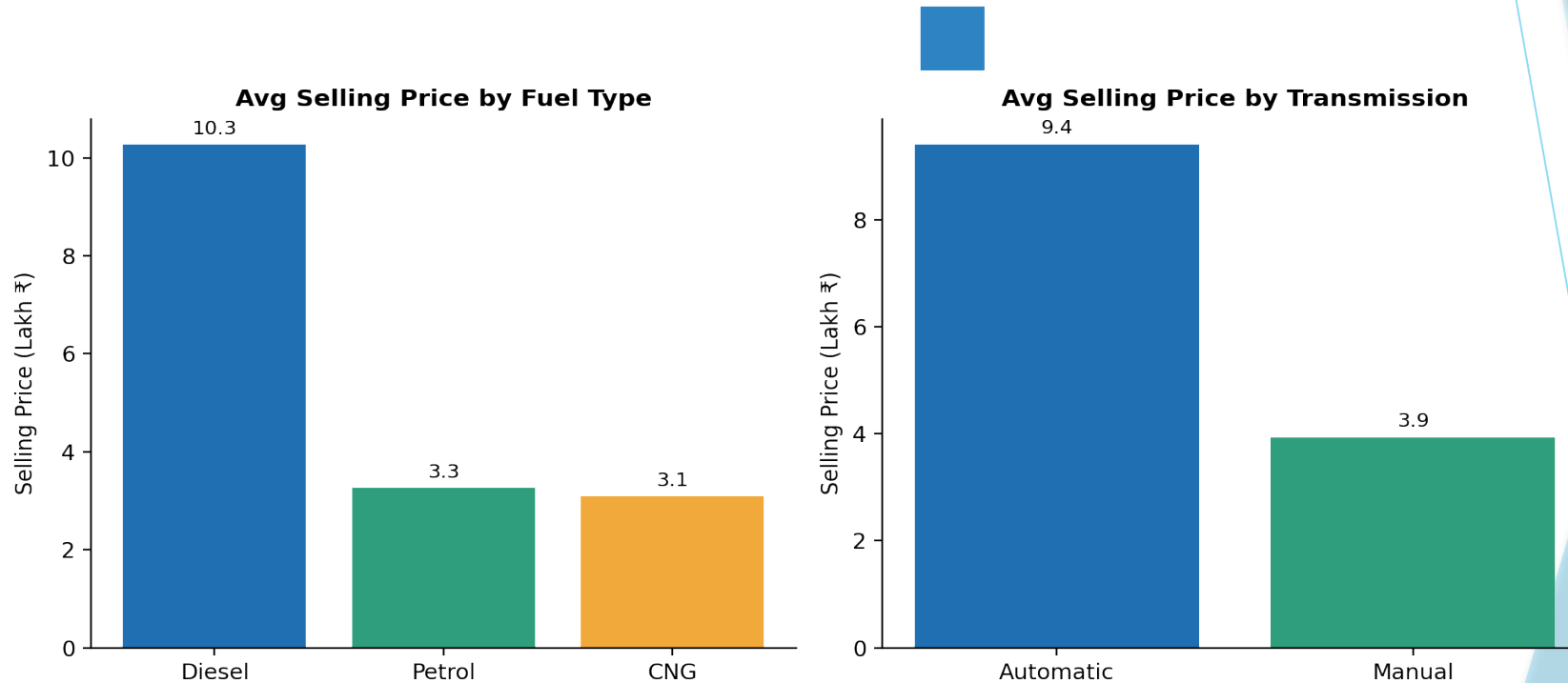
# Technology Used

- Python (Pandas, NumPy) – data cleaning & analysis
- Matplotlib & Seaborn – data visualization
- Jupyter Notebook – exploratory data analysis
- CSV Dataset – CarDekho used car listings (301 records, 9 features)
- Git & GitHub – version control and submission



# RESULTS

## Average Selling Price by Fuel Type & Transmission

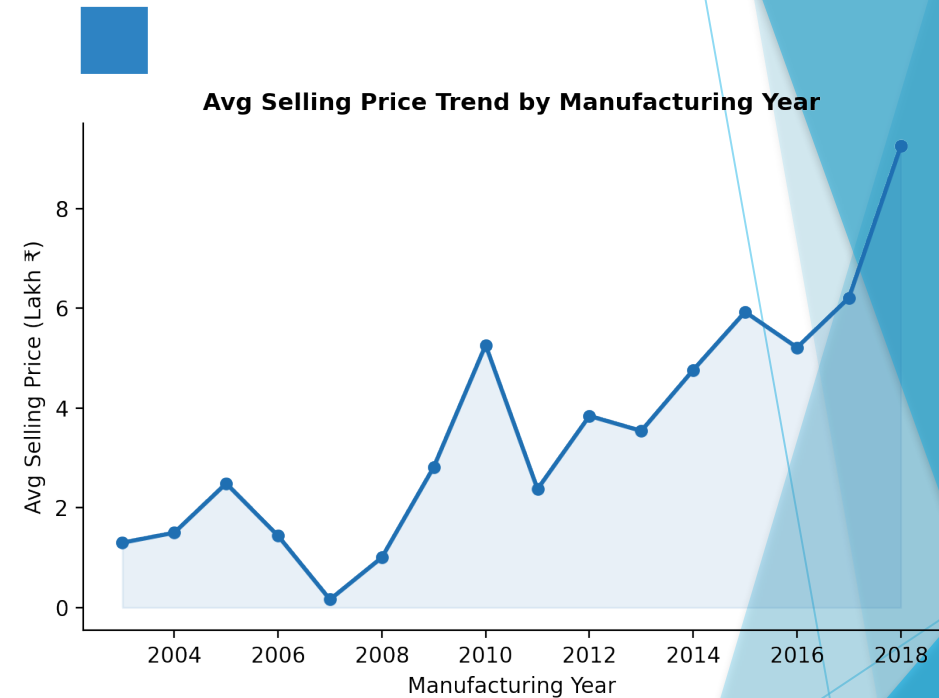
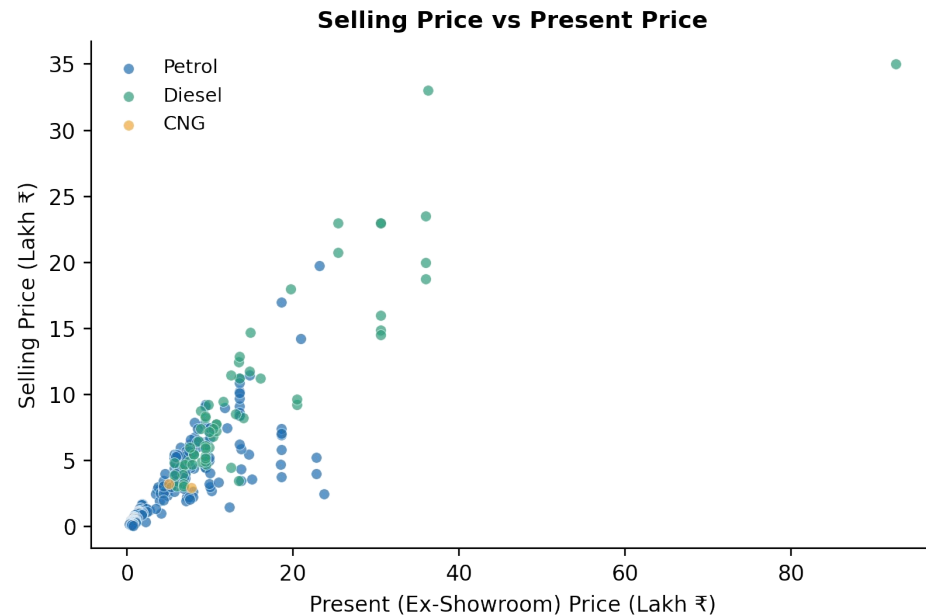


*Insight: Diesel and automatic-transmission cars fetch roughly 2-3x higher average resale prices than petrol/manual cars.*

*Source: CarDekho Used Car Price dataset (Kaggle) – 301 listings, 2003-2018*

# RESULTS

## Price Relationships & Year-wise Trend



*Insight: Selling price closely tracks present price ( $r \approx 0.88$ ), the strongest predictor of resale value.*

*Newer cars (post-2015) fetch higher prices from lower mileage and less depreciation.*

*Source: CarDekho Used Car Price dataset (Kaggle) - 301 listings, 2003-2018*

# Thank you

**GitHub:** [github.com/Vishalavalige/car-market-trends-analysis](https://github.com/Vishalavalige/car-market-trends-analysis)

